

Face Liveness Detection and Facial Recognition

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Abstract—With the more extensive utilization of sophisticated spoofing attacks, effective face liveness detection and recognition systems are becoming the order of the day to secure identity verification in real-time. This system combines OpenCV, Convolutional Neural Networks (CNNs), and FaceNet for authenticating accurately by identifying the user's identity and liveness. Live webcam frames are captured and transmitted to the server for analysis using OpenCV which handles the frame extracting and preprocessing pipeline. At the same time, HTML5 getUserMedia() method enables video capture from the client's device in real time. With the help of texturing and microexpression through CNN, a user can be distinguished from a fake attack or other impersonations. This paper talks about the performance of FaceNet as an example of a deep-learning facial recognition software that increases security through accurate facial embedding verifications. This work analyzes the system's efficacy in various settings, including its sophistication in dealing with advanced spoofing. The approach aims to achieve secure, near real-time face liveness detection and recognition, meeting the high reliability required by modern authentication systems.

Index Terms—Face Liveness Detection, Face Recognition, OpenCV, HTML5 webcam capture, Convolutional Neural Network (CNN), FaceNet, Real-Time Authentication, Spoofing Detection, Biometric Security, Identity Verification.

I. INTRODUCTION

Face liveness detection and facial recognition are now firmly established as critical security technologies; they are being deployed to mitigate increasingly sophisticated threats from spoofing strategies involving photo attacks, video replays, and AI-generated deepfakes. Inference on face-alone systems (without behavioral biometric traits) is not robust against unauthorized access using the traditional methods. Static images or recordings can beat conventional recognition techniques. The challenge, of course, is to verify the user's identity and that the detected face is live. This helps keep applications like banking, mobile authentication, and access control safe by ensuring that identity verification is accurate and dependable.

New developments in artificial intelligence (AI) and real-time data processing show great promise in addressing these issues. Tools such as OpenCV help with image processing,

while the HTML5 getUserMedia() method allows live video capture. Convolutional Neural Networks (CNNs) play a role in checking if the user is genuinely present. By utilizing FaceNet for recognition, we can create a trustworthy system to confirm a user's identity and authenticity. The getUserMedia() method captures live video feeds needed for real-time facial cue detection, and OpenCV helps extract and prepare frames effectively. The CNN-based liveness model can pick up subtle facial details.

II. MOTIVATION OF PROJECT

As we all know, facial recognition technology has been widely used nowadays for secure authentication, which has made it important to be resistant to phishing attacks. The ability to shoot images, videos, or even objects, along with the depth of AI that can even fool traditional facial verification, highlights the need to verify not only a person's identity but also the "life" of their face. This is to ensure that the detected face is a real body and has not been spoofed.

This research study is motivated by the increasing need for a secure, reliable, user-friendly solution for detecting and recognizing facial images. The system aims to improve the security of facial authentication systems by providing liveness detection and ensuring that authorized persons are attempting to connect through secure, dependable networks and have access to relevant data.

The system also intends to unlock the potential of using this combined powerful technology in various applications by utilizing OpenCV for image processing, HTML5 getUserMedia() for real-time video streaming, CNNs for liveness detection, and FaceNet for accurate identification.

III. MACHINE LEARNING (ML)

Machine Learning (ML), being a branch of AI, allows systems to learn from data structures and make decisions or predictions without an explicitly programmed system. For facial image detection and recognition, ML is widely used.

Using techniques such as supervised learning, ML models are trained on labeled datasets of live and spoof models to classify and help identify and mitigate security risks. These models analyze multiple phenomena, including facial movements and gestures. Differentiating between live faces and still images or video is one example.

Accordingly, this study aims to achieve safe, trustful, and real-time detection and recognition functionalities through the enablement of an ML-assisted infrastructure that facilitates the secure and reliable hosting of applications under a multi-trust paradigm.

IV. DEEP LEARNING (DL)

Deep learning is now essential for improving biometric systems, especially when telling if a face is real. These deep learning tools help make face recognition technology more accurate and secure. These systems are critical because they are vital to confirm a user's identity and protect sensitive information, such as in banking, secure entry systems, and personal devices.

Convolutional Neural Networks, being excellent with visual data, become an ideal choice for face liveness detection. They analyze facial details to recognize if the face presented is a genuine human being or a reproduction such as a photograph, video, or mask. When these networks are taught using many labeled images, they learn to see the small details and patterns in videos that help distinguish a real face from a fake one. This ability to recognize if a face is genuine is essential for ensuring that face recognition systems can't be easily tricked, which is a big worry for security.

Using deep learning in systems that check if a face is real and identify people can make them work much better. It helps to prevent mistakes, like saying someone exists when they don't and missing out on real people. This helps to verify identities more easily. These innovative methods also keep the platform safe from dangers, making users feel secure and trust it more. Through the application of deep learning, the platform makes identity verification secure and accurate, and it also improves the overall experience for users. This renders it a useful tool in a variety of real-world scenarios where identity checks and security are vital.

V. RELATED WORK

Schroff et al. [1] presented FaceNet, a method using deep learning to create a single type of representation for recognizing and organizing faces. This approach has dramatically helped face recognition systems work better. It also lays a solid groundwork for detecting if a face is real or fake by making sure the facial representations are accurate, which helps tell apart real faces from fakes.

Yi Sun et al. [2] explored how deep learning methods, especially CNNs, can be used for recognizing faces. This research demonstrated how deep learning improves face recognition accuracy and strengthens the system against attacks, which is crucial for dependable verification.

R. K. Soni et al. [3] discussed a method for face liveness detection in recognition systems. Using CNNs, they demonstrated that deep learning can effectively distinguish real faces from spoofed attempts, making biometric authentication systems more secure and reliable.

R. Singh et al. [4] reviewed various methods for detecting liveness in face recognition, covering both image-based and motion-based techniques.

S. S. Vatsa et al. [5] discussed deep learning techniques and the use of CNNs in detecting spoofing attempts which is a concern in face recognition systems.

R. Koshi et al. [6] Discusses entropy based deep CNN architectures and improved spoof detection accuracy using optimized models for face liveness detection.

Y. Taigman et al. [7] proposed DeepFace, a deep model that achieved human-level performance in face verification. They used convolutional neural networks to improve facial feature understanding, enhancing the detection of live faces.

A. T. Nguyen et al. [8] suggested a system of face recognition and spoof detection via deep learning. Their work proved that CNNs are able to effectively detect spoof attacks with high recognition accuracy.

M. Nogueira et al. [9] discussed different methods of spoof attack prevention in face recognition. They discussed deep learning methods, such as CNNs and GANs, for enhancing biometric security.

J. Liu et al. [10] discussed the use of deep convolutional networks in improving real-time face recognition accuracy and how deep learning models prove to be efficient in handling facial information.

H. Zhao et al. [11] reviewed different liveness detection techniques for face recognition systems, focusing on traditional and deep learning approaches.

Z. Chen et al. [12] proposed using CNNs for detecting liveness in face recognition systems. Their study demonstrated that deep learning models can effectively differentiate between live faces and spoofing attempts, enhancing biometric security.

A. C. C. Yip et al. [13] surveyed techniques of spoofing in face recognition systems that focus on the use of deep learning in counteracting such attacks. The paper provided a roadmap for future research on improving the robustness of liveness detection systems.

H. Nguyen et al. [14] explored a multi-modal approach to liveness detection using deep learning. The paper integrated multiple data sources (e.g., RGB images and depth information) to enhance the accuracy of face liveness detection systems.

R.T.L. Uren et al. [15] investigated liveness detection and face recognition using a multi-task deep learning model. The study showed improved accuracy and efficiency through joint training.

Haoliang Li et al. [16] Utilized an unsupervised domain adaption method using adversarial learning in order to align feature distributions between source and target domains for the purpose of face anti spoofing. Cross-dataset generalization was achieved even without target domain data.

Zhi Li et al. [17] Used a teacher-student architecture to create compact representations that were sturdy against unseen attack types. An one-class knowledge distillation framework was used for presentation attack detection.

VI. DATASET DETAILS

The ROSE-YOUTU Face Liveness Detection Database [16] is a large dataset with its key motive in assisting face anti-spoofing studies. It records rich illumination conditions, models of cameras, and attack types to improve model generalizability [17]. There are 4225 videos taken from 25 subjects out of which 3350 videos from 20 subjects (5.45 GB) are available. There are between 150 and 200 video clips per subject in the dataset, and each clip takes around 10 seconds. They were recorded from five devices:

- Hasee smartphone (640 × 480)
- Huawei smartphone (640 × 480)
- iPad 4 (640 × 480)
- iPhone 5s (1280 × 720)
- ZTE smartphone (1280 × 720)



Fig. 1. Example images from the dataset.

A. Spoofing Attacks

The dataset considers three major types of spoofing attacks:

- **Printed Paper Attack:** A still image printed on A4 size paper.
- **Video Replay Attack:** A face video replayed on a Lenovo LCD or Mac screen.
- **Mask Attack:** Paper masks with cut-out sections for the eyes and mouth.

TABLE I
DATASET STRUCTURE AND DIMENSIONS

Category	Details
Resolution	640×480 / 1280×720
Total Videos	4225 videos (3350 public)
Subjects	25 total (20 public)
Devices Used	Hases, Huawei, iPad 4, ZTE
Genuine Videos	25 per device per subject (<5)
Spoof Attacks	Printed Paper, Video Replay

VII. PROPOSED ARCHITECTURE

The new system addresses important security concerns using biometrics, like making sure users are real, confirming their identity, and keeping data safe. It uses strong technology powered by AI and combines tools like HTML5 webcam capture, OpenCV, Convolutional Neural Networks, and FaceNet models to create a smooth and safe way to verify users. The goal is to stop fake attempts and make sure that only real users can get in.

In this system, users are identified through their faces. When they initiate the process, they experience three primary steps: Video Capture, Face Liveness Detection, and Facial Recognition. All these steps combine to create a secure and convenient system for verifying identities based on biometrics.

A. Video Capture Module

It all starts with capturing video in real time through HTML5 getUserMedia(). This lets the system grab the video feed right from the user's device. To make sure the video frames are ready for what's next, OpenCV comes into play to tweak them. This helps guarantee that the captured video is good enough for accurate liveness detection and recognition.

B. Face Liveness Detection Module

This part of the system leverages a Convolutional Neural Network (CNN) to distinguish real faces from fake ones, like pictures or videos. It has learned from a mix of genuine and fake samples, which helps it recognize live faces effectively.

When someone shows their face, the system checks if it's real. Along with passive liveness detection, an active liveness detection functionality has also been incorporated. The user is prompted to perform a random sequence of facial actions which includes blinking, nodding, and smiling. This is detected using Mediapipe's Facemesh and if the user fails to perform it within the given time frame, the user is classified as spoof.

- **Live:** The facial recognition module recognizes the live face and passes it on.
- **Bluffing:** The face is identified as spoof and declined by the system to avoid unauthorized access.

When the liveness check is confirmed, the system moves on to facial recognition, taking a unique image of the user's face with FaceNet. This photograph is compared with biometric information stored in a secure database to identify the individual. This means that these personal details can be secured to only registered users.

C. Authentication Outcomes

Based on the results of the facial recognition, the system provides two outcomes:

- **Authenticated:** The system successfully logs in the user when it recognizes their face and finds a match in its records.
- **Authentication Failed:** If the system does not find a matching face, it will not allow access.

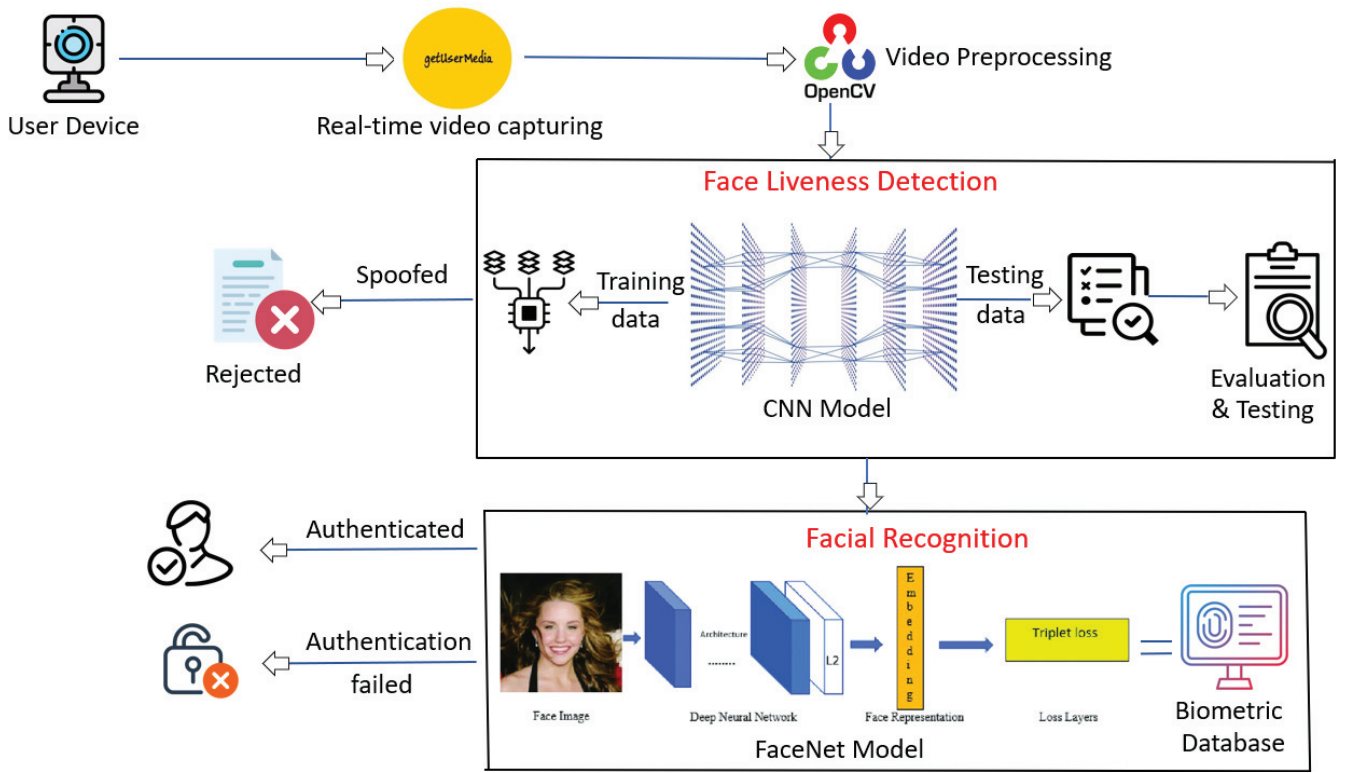


Fig. 2. Proposed System Architecture.

VIII. PROPOSED ALGORITHM

A. Convolutional Neural Network (CNN)

CNNs are ideal for automatically learning distinguishing features in facial images. The architecture of CNNs includes several layers, each contributing to the learning process. These layers include convolutional layers, activation layers, pooling layers, and fully connected layers. The following outlines the CNN algorithm for face liveness detection and recognition.

1) *Convolution Layer*: The convolution layer is the main part of a CNN. For face liveness detection, these layers find facial features and recognize patterns like skin texture or the shape of the face. The convolution process works like a sliding window, where a filter moves across the image. It calculates how well the filter matches the image area it covers. This creates a feature map that shows where certain features are found.

Mathematically, for an input image I of size $H \times W$, and a filter K of size $k_h \times k_w$, the output feature map $O(I, j)$ at position (I, j) is calculated as:

$$O(i, j) = \sum_m \sum_n [(i + m, j + n) \cdot K(m, n)] \quad (1)$$

Multiple filters are applied to capture different facial features (such as edges, textures, or spots), which are then combined into a more complex representation of the face.

2) *Activation Function (ReLU)*: After each convolution step, we use an activation function to add some non-linearity to the network. This helps it understand more complicated

patterns. The most popular activation function in CNNs is called ReLU, which stands for Rectified Linear Unit. What ReLU does is simple: if the input is positive, it keeps that value; if it's not, it changes it to zero. We can write this as:

$$f(x) = \max(0, x) \quad (2)$$

By using ReLU, the network can pay attention to the important features and ignore negative values. This makes learning easier and speeds up the learning process.

3) *Pooling Layer*: CNNs use a pooling layer to make the feature map smaller while keeping the important information. This layer helps by reducing the detail in the feature maps, which makes the model run better and helps prevent overfitting. The most common way to pool is called max pooling. It picks the highest value from each area of the feature map, ensuring that the main features stay. For instance, when we use max pooling with a 2×2 window on a feature map, we take the highest value from each 2×2 section, cutting the size of the feature map in half.

4) *Fully Connected Layer*: After going through several layers of convolution and pooling, we send the features to fully connected layers. These layers take the features we've extracted and learn how to match them to specific results, like whether the face is real or fake and what the identity is. In these fully connected layers, each neuron links to every neuron from the previous layer. The output from these layers goes into a softmax function, which changes the raw scores into

probabilities for each category. The category with the highest probability becomes our prediction, telling us if it's a live face or a spoofed one. In simple math terms, if we have a fully connected layer with an input vector X , a matrix of weights W , and a vector of biases b , we find the output y like this:

$$Y = W \cdot X + b \quad (3)$$

Then, the softmax function helps us turn that output into probabilities:

$$P(\text{liveness}) = P(y_i) = e^{x_i} / \sum_j e^{x_j} \quad (4)$$

Here, $P(\text{liveness})$ shows the chances that the face we're looking at is live.

B. Model Training

Once the CNN architecture is defined, the model is trained using a labeled dataset. For face liveness detection, the labels will be either live or spoof (e.g., from a photo or video). In face recognition, each label represents a different person.

IX. RESULTS AND DISCUSSION

A. Preprocessing and Dataset Preparation

To ensure high-quality input for training, the following preprocessing steps were applied:

- **Frame Extraction:** The first frame of each video was selected for classification.
- **Resizing:** Images were resized to **224×224 pixels** to match the input size of ResNet-18.
- **Data Augmentation:**
 - Real images were augmented with *random rotations, brightness/contrast adjustments, and flipping*.
 - Spoof images were resized and normalized but not augmented.
- **Normalization:** Pixel values were scaled to the range $[-1, 1]$ using $\text{mean}=[0.5, 0.5, 0.5]$ and $\text{std}=[0.5, 0.5, 0.5]$.

B. ResNet-18 Model Architecture

The proposed system utilized **ResNet-18**, a deep residual network designed to overcome the vanishing gradient problem in deep networks. The architecture consists of:

- **Initial Convolution and BatchNorm layers.**
- **Four Residual Blocks**, each containing:
 - Two convolutional layers (3×3 filters).
 - Batch normalization and ReLU activation.
 - Identity shortcut connections.
- **Fully Connected Layer:**
 - Outputs a binary classification (Real/Spoof).
 - A dropout layer (0.5) was added before the final classification layer to reduce overfitting.

TABLE II
EPOCH-WISE TRAINING RESULTS

Epoch	Train Loss	Val Loss	Val Accuracy
1	0.0265	0.0342	97.48%
2	0.0102	0.0016	99.96%
3	0.0077	0.0074	99.48%
4	0.0071	0.0081	99.54%
5	0.0054	0.0023	99.87%
6	0.0018	0.0012	99.96%
7	0.0025	0.0011	99.97%
8	0.0025	0.0016	99.95%
9	0.0009	0.0011	99.96%
10	0.0015	0.0019	99.98%

TABLE III
PERFORMANCE METRICS OF RESNET-18 MODEL

Metric	Value
Accuracy	99.5%
Precision	98.8%
Recall	99.1%
F1-Score	98.9%

C. Training Details

The model was trained with the following settings:

- **Loss Function:** Weighted Cross-Entropy Loss (real: 1.0, spoof: 3.0) to handle class imbalance.
- **Optimizer:** Adam (lr=0.0005).
- **Scheduler:** Learning rate reduction on validation loss plateau.
- **Batch Size:** 64.
- **Training Strategy:**
 - 80% of the dataset for training, 20% for validation.
 - Weighted sampling to balance real and spoof classes.
 - Early stopping after 5 epochs of no improvement.

D. Epoch-wise Training Results

Table II presents the epoch-wise training results.

E. Performance Evaluation

The performance metrics of the proposed model are shown in Table III.

F. Comparison with Existing Methods

Table IV provides a comparison of the proposed method with existing face-liveness detection techniques.

G. Limitations and Future Work

Despite the high accuracy, some challenges remain:

- Performance drops under extreme low-light conditions.
- Real-time inference may require optimization for low-power devices.
- Future work will focus on incorporating multimodal biometric verification and infrared-based liveness detection to improve robustness.

TABLE IV
COMPARISON OF FACE LIVENESS DETECTION METHODS

Method	Accuracy (%)	False Positives (%)
Traditional CNN	92.5	4.5
MobileNetV2	96.7	3.2
ResNet-18 (Ours)	99.5	1.1

X. OUTCOMES

The results of this AI-powered face liveness detection and recognition project will be:

- Improved the security of biometric systems by accurately identifying real/fake faces to prevent unauthorized access.
- It can help by quickly and accurately recognizing faces for things like mobile phones and online logins.
- Real-time face verification that facilitates quicker and better identity verification for sectors like banking, health-care, and e-commerce.
- Minimizes the chances of attackers stealing identity and committing fraud by identifying spoofing techniques like masked attacks, replay attacks, printed attacks and facilitates increased trust.

XI. CONCLUSION

In this project, a face liveness detection and recognition system was implemented that uses Convolutional Neural Networks (CNNs), which is a deep learning technique. Our system distinguishes real faces from fake faces, therefore increasing the reliability of face recognition technology. In fast loads analysis and identification of faces, this system protects against tricks, including photos, videos, and masks, which will fool security systems. The system improves the security of biometric systems, and also enhances user experience via fast, accurate, and reliable performance. The application of the system is in many fields, such as finance, health care, and online shopping, where securely and reliably confirming identity is important. Its successful application shows that deep learning is becoming more valuable for security and confirming users, setting the stage for better accuracy and easier usage in the future

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